

## Chapter 2 Review

### Learning Objectives

- Consolidate understanding of all data structures: lists, dictionaries, tuples, and sets
- Solve 综合 problems that require selecting and combining multiple data structures
- Evaluate design decisions and justify data structure choices for given problems

### Engage (5-7 minutes)

Over the past eleven lessons, you have mastered the complete toolkit of Python data structures. Each has strengths and weaknesses, and knowing when to use which is what separates a good programmer from a great one. Today we consolidate this knowledge through challenging problems that require thoughtful selection and combination of all data structures.

### Explore (10-12 minutes)

Take this problem: “Given a list of words, group them by their sorted letter pattern (anagrams share the same sorted form).” Think about which data structure groups items by a common key—a dictionary. Now solve: “Find all pairs of numbers in a list that sum to 100”—which structure gives  $O(1)$  lookup for complements—a set. Discuss with a partner: why did you choose each structure? What would happen if you used the other?

### Explain (10-12 minutes)

This chapter covered four fundamental data structures. Lists provide ordered, mutable sequences for collections accessed by position. Dictionaries offer key-value mapping for fast lookups and structured data. Tuples deliver immutable, ordered data for fixed records and safe returns. Sets enable uniqueness checking and mathematical operations. The key skill is matching problem requirements to data structure capabilities: order matters use lists, fast lookup use dictionaries, immutability needed use tuples, uniqueness critical use sets. Real-world programs often combine all four in a single solution.

### Elaborate (8-10 minutes)

Complete this comprehensive challenge: build a word frequency analyzer that reads a text file, counts word frequencies using a dictionary, identifies anagrams using sorted tuples as dictionary keys, finds words that appear only once using set operations, and stores results in a well-organized dictionary. Then create a contact book that supports search by name, search by phone, group contacts by category, and export the data as a formatted report. Test with real-world text and datasets of varying sizes.

### Evaluate (5-8 minutes)

For each scenario, identify the best primary data structure and justify your choice: (1) storing playlist songs that can be reordered, (2) caching website configuration, (3) tracking unique visitors to a website, (4) mapping country codes to country names, (5) storing GPS coordinates of visited locations. Then write a reflection on your learning journey through this chapter and set three goals for the next chapter.